

RUBAB ZAHRA SARFRAZ

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EDUCATION

University of Illinois Chicago (UIC) <i>Ph.D. in Computer Science</i> Advisor: Boris Glavic	2025 – present Chicago, IL
Lahore University of Management Sciences (LUMS) <i>M.S. in Computer Science</i> Advisor: Ihsan Ayyub Qazi	2016 – 2018 Lahore, Pakistan
University of Engineering & Technology (UET) <i>B.Sc. in Computer Engineering (with honors)</i> Advisor: Irfan Ullah Chaudhary	2012 – 2016 Lahore, Pakistan

PUBLICATIONS

- Rubab Zahra Sarfraz**, Boris Glavic. “Speeding Up Transformation Search with Lightweight Statistics”. In *4th International Workshop on Tabular Data Analysis (TaDA) at VLDB 2026*.
- Rubab Zahra Sarfraz**, Samar Haider. “Vizard: Improving Visual Data Literacy with Large Language Models”. In *7th International Workshop on Big Data Visual Exploration and Analytics (BigVis) at VLDB 2024*. [PDF]
- Rubab Zahra Sarfraz**. “Towards Semi-Supervised Data Quality Detection in Graphs”. In *13th International Workshop on Quality in Databases (QDB) at VLDB 2024*. [PDF]
- Nida Munawar, **Rubab Zahra Sarfraz**, Maria Costello, David Robinson, Colm Bergin, Elaine Greene. “Risk Factors and Outcomes of Delirium in Hospitalized Older Adults with COVID-19: A Systematic Review and Meta-Analysis”. In *Aging and Health Research*, 2023.

RESEARCH EXPERIENCE

University of Illinois Chicago (UIC) <i>Mentor: Boris Glavic</i> Working on scalable transformation discovery methods for data integration in heterogeneous data lakes.	Aug. 2025 – present Chicago, IL
Lahore University of Management Sciences (LUMS) <i>Mentor: Ihsan Ayyub Qazi</i> Studied misinformation in the Global South through algorithmic audits of Facebook’s content moderation policies, evaluation of fake news detection tools in local contexts, and large-scale surveys of news consumption habits.	Oct. 2017 – Jun. 2018 Lahore, Pakistan
University of Engineering & Technology (UET) <i>Mentor: Irfan Ullah Chaudhary</i> Designed and built a network monitoring plugin for OpenStack Monasca using IP tables and Grafana to improve visibility and mitigate DDoS threats in virtualized environments.	Jun. 2015 – Jun. 2016 Lahore, Pakistan

AWARDS & HONORS

Invited Participant, IVLP, U.S. Department of State	2024
Finalist, USAID FDI Grant of \$100K (Top 5/50+ Applicants)	2024
Finalist, U.K. Climate Finance Accelerator (Top 10/100+ Applicants)	2023
Runner-Up (2/40), CERN Openlab Lightning Talks (Program Acceptance Rate: 2%)	2018
Diversity Scholar, KubeCon by the Linux Foundation (First ever Scholar selected from Pakistan)	2018
Dean’s Honor List, University of Engineering & Technology (Top 10% of class)	2012 – 2016
Rank 3 out of 105,000 students, Gujranwala High School District	2012

WORK EXPERIENCE

BridgeLinx

Nov. 2022 – May 2025

Data Lead

Lahore, Pakistan

- Established the company's data infrastructure from the ground up, utilizing Snowflake for data lakes, Prefect for quality-driven data pipelines, and Tableau/Metabase for front-end, adopted by **40%** of the workforce.
- Built a temporal geospatial data pipeline, boosting customer acquisition by **25%** across **150+ transportation lanes** through optimized operational reach and resource allocation.
- Incorporated data quality checklist in product sprints. Created and deployed a data quality bot, improving data completeness from **45% to 98%**, resulting in 30% faster decision-making across **9 teams** and **200+ data points**.
- Engineered a real-time bidding system employing advanced analytics to optimize Return on Capital (ROC) through predictive client settlement behavior analysis upon order booking.

Lahore University of Management Sciences (LUMS)

Jan. 2025 – May 2025

Adjunct Faculty

Lahore, Pakistan

- Designed and taught an advanced "Data Engineering for AI Systems" course for MS in AI students at LUMS, covering end-to-end data pipelines, from ingestion and processing to analytics and deployment, through interactive labs and real-world assignments.
- Effectively mentored a diverse cohort (including non-technical students), ensuring inclusive learning by providing tailored support, hands-on projects, and practical industry-focused best practices for comprehensive skill development.

Finja

Jan. 2020 – Nov. 2022

Data Lead

Lahore, Pakistan

- Researched the low-resourced, undocumented SME market, engineered a machine learning pipeline identifying 40+ predictive variables, reducing loan approval time by **35%**, while keeping the **NPLs below 0.5%** with **\$25M in issuance**.
- Advocated for and expanded the ML model's business evaluation metric from overall to category-wise NPLs to protect vulnerable MSMEs from market-driven penalties, using a real-time feedback loop for classifying bad loans into categories.
- Led a multidisciplinary team to launch Pakistan's first data-driven **SECP-approved** P2P lending, investment and payment **engine**, resulting in **\$1M+** lending issuance to **3000+ retail stores**.
- Led a 10-member data science team, achieving a **40%** reduction in data processing time and transitioning 8 departments to fully adopt data-driven decision-making, increasing operational efficiency by **25%**.

Oct. 2018 – Dec. 2019

Data Engineer

Lahore, Pakistan

- Architected a scalable high-quality data architecture for financial transactions (lending, payments, investments), optimizing for performance, security, and compliance across a multi-service platform serving **300K customers**.
- Built and managed a high-availability data lake and hybrid data pipelines, integrating real-time streaming and batch processing to prioritize workloads efficiently across **50+ dashboards**, **8 departments** and **3 products**.
- Developed a high-accuracy sanctions screening system in-house for KYC, enabling real-time fraud detection on incoming wallets and saving **\$150K**. Successfully identified **two high-risk matches**, mitigating compliance risks and preventing financial losses.
- Enhanced search engine relevance for a B2C marketplace (from **40%** to **80%** relevance ratio) by improving vendor product data quality and using machine learning models in an adaptive pipeline with human-in-the-loop validation.

CERN

Jul. 2018 – Aug. 2018

Intern, Software Engineer

Geneva, Switzerland

- Spearheaded the test deployment **Ceph clusters with Rook on Kubernetes**, leveraging both OpenStack VMs and Ironic hosts.
- Defined and implemented robust evaluation metrics, demonstrating the superior latency and user-friendliness of the new cluster deployment approach, reducing setup time significantly.
- Enhanced the Orchestrator CLI with **RGW support**, enabling rapid S3 service provisioning in just **1-2 minutes**.
- Published a **blog post** with the findings in collaboration with CERN and Ceph team.

RedHat (via Outreachy)

Dec. 2017 – Mar. 2018

Intern, Software Engineer

Remote

- Improved distributed cluster management by adding performance dashboards. Pull requests: [1] [2] [3]
- Converted the decentralized architecture of Ceph Manager to a **centralized one** for improved coherence with the codebase.

Meta

Feb. 2017 – Mar. 2017

Mentee, Software Engineer

Remote

- Extended the Osquery open source project by implementing a **virtual table in C++ for listing Python packages** installed on a server.
- This functionality was actively used for securing Meta's data center servers against **vulnerabilities introduced by PyPI in 2017**.

TEACHING EXPERIENCE

Teaching Assistant , Database Systems University of Illinois Chicago (UIC) <i>Instructor: Boris Glavic</i>	Spring 2026
Teaching Assistant , Programming Practicum University of Illinois Chicago (UIC) <i>Instructor: Scott Reckinger and Mark Hodges</i>	Fall 2025
Adjunct Faculty , Data Engineering for AI Systems Lahore University of Management Sciences (LUMS)	Spring 2025
Instructor , Introduction to Data Science Lahore School of Economics (LSE)	Summer 2023
Teaching Assistant , Advanced Operating Systems Lahore University of Management Sciences (LUMS) <i>Instructor: Muhammad Hamad Alizai</i>	Spring 2017
Teaching Assistant , Programming Fundamentals University of Engineering & Technology (UET) <i>Instructor: Irfan Ullah Chaudhary</i>	Spring 2016

COMMUNITY INVOLVEMENT

Reviewer, IEEE VIS Workshop on Visualization for AI Explainability (VISxAI)	2024
Board Member, Computer Science Alumni Network , LUMS	2024
Advisor, Securities and Exchange Commission of Pakistan (for drafting regulations for P2P lending)	2022
Open Source Contributor, Ceph Blog: Evaluating Ceph Deployments with Rook	2018
Open Source Contributor, Docker Docs Hackathon . Pull requests: [1] [2] [3] [4]	2017

TALKS

“Elevating Trust in Your Data with Python” , PyCon Pakistan	2024
“Reshaping the Financial Sector with Artificial Intelligence”, Information Technology University	2022
“Debugging the Startup Space: Tech Careers”, LUMS Women in Computing	2022
“Best Practices in Operationalizing Machine Learning” , AWS Startup CTO Forum	2022
“Breaking the Bias”, Systems Limited	2021
“How Technology can be Your Best Friend”, Meta Developers Circle	2020
“Evaluating Ceph Deployment with Rook” , CERN	2018

TECHNICAL SKILLS

Expertise: Data Engineering, Data Architecture, Data Modeling, Data Visualizations, Data Product Management

Languages & Scripts: Python, C++, SQL, HTML, $\text{\LaTeX}$

Machine Learning: Scikit-learn, PyTorch, Transformers

Data Lake & Quality: BigQuery, Snowflake, S3, Great Expectations, Pandera

Data Visualization: Tableau, Looker Studio, Streamlit

Data Infrastructure: GCP (Cloud Functions, Storage, Dataflow, SQL), AWS (SageMaker, Lambda), Prefect, FastAPI, Flask, Docker